

8. Limit Theorems.

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8.1 Modes of Convergence

Definition 8.1. Let E be a Borel subset of $\mathbb{R}^n$ and let (f_n) be a sequence of functions where each $f_n \in L^p(E)$, $p \geq 1$.

- (i) $f_n \rightarrow f$ *uniformly on E* if for any $\epsilon > 0$, $\exists N = N(\epsilon)$ such that, for all $n \geq N$,

$$\|f_n - f\|_\infty = \sup_{x \in E} (|f_n(x) - f(x)|) < \epsilon.$$

(Note that we need $f_n \in L^\infty(E)$ for the sup to be finite in general.)

- (ii) $f_n \rightarrow f$ *pointwise on E* if for each $x \in E$, given $\epsilon > 0$, there exists $N = N(\epsilon, x)$ such that $|f_n(x) - f(x)| < \epsilon$ for all $n \geq N$.
- (iii) $f_n \rightarrow f$ *almost everywhere (a.e.) on E* if there is a null set $F \subseteq E$ such that $f_n \rightarrow f$ pointwise on $E \setminus F$.
- (iv) $f_n \rightarrow f$ *in L^p -norm (in the p -th mean)* if $\|f_n - f\|_p \rightarrow 0$ as $n \rightarrow \infty$ i.e. for any $\epsilon > 0$, $\exists N = N(\epsilon)$ such that

$$\|f_n - f\|_p \triangleq \left(\int_E |f_n - f|^p \, dm \right)^{\frac{1}{p}} < \epsilon$$

for all $n \geq N$.

Theorem 8.2. With (f_n) as above, the only valid implications are the following: (i) $\implies$ (ii) $\implies$ (iii). For finite measures, (i) $\implies$ (iv).

CruX. • (i) $\implies$ (ii): For $n \geq N(\epsilon)$, $|f_n(x) - f(x)| \leq \|f_n - f\|_\infty < \epsilon$.

- (ii) $\implies$ (iii): Since $f_n \rightarrow f$ pointwise everywhere on E , the required null set is $F = \emptyset$.

- (i) $\implies$ (iv) for finite measures:

$$\|f_n - f\|_p = \left(\int_E |f_n - f|^p \, dm \right)^{1/p} \leq \|f_n - f\|_\infty m(E)^{1/p}.$$

Since $m(E) < \infty$, the right side vanishes as $n \rightarrow \infty$.

Let $E = [0, 1]$. Taking $f_n = \mathbb{I}_{(0, 1/n]}$ shows (ii) $\not\Rightarrow$ (i).

(iii) $\not\Rightarrow$ (ii) is conceptually immediate: (ii) strictly requires pointwise convergence on the entire set E , leaving no scope for failure, whereas (iii) allows failure on any $F \subseteq E$ with $m(F) = 0$.

(iii) $\not\Rightarrow$ (iv) is seen using $f_n = n\mathbb{I}_{(0, 1/n]}$. The sequence converges pointwise to 0 everywhere, but $\|f_n - 0\|_p^p = n^p(1/n) = n^{p-1}$. For $p \geq 1$, this does not converge to 0, so it fails in L^p .

Finally, (iv) $\not\Rightarrow$ (iii) can be seen using the *sliding typewriter* example. Any integer $n \geq 1$ can be uniquely written as:

$$n = 2^k + j \quad \text{where } k \geq 0 \text{ and } 0 \leq j < 2^k$$

We define $f_n(x)$ as the indicator function of the shifting interval I_n :

$$f_n(x) = \mathbf{1}_{I_n}(x) = \begin{cases} 1 & \text{if } x \in \left[\frac{j}{2^k}, \frac{j+1}{2^k} \right] \\ 0 & \text{otherwise} \end{cases}$$

Here, $\|f_n\|_p^p = 1/2^k \rightarrow 0$ as $n \rightarrow \infty$, proving L^p convergence. However, for any fixed $x \in [0, 1]$, $f_n(x)$ oscillates infinitely between 0 and 1, failing to converge pointwise anywhere. $\square$

8.2 Probability

Definition 8.5. We say that random variables $(X_n)_{n \geq 1}$ are independent if for any $k \in \mathbb{N}$ the variables $X_1, X_2, \dots, X_k$ are independent (see Definition 3.18).

Definition 8.6 The sequence (X_n) converges to X *in probability* if for each $\epsilon > 0$,

$$P(|X_n - X| > \epsilon) \rightarrow 0$$

as $n \rightarrow \infty$.

Lemma 8.7. *The following conditions are equivalent:*

(i) $Y_n \rightarrow 0$ *almost surely*,

(ii) *for each* $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} P \left(\bigcup_{n=N}^{\infty} \{\omega : |Y_n(\omega)| \geq \epsilon\} \right) = 0.$$

CruX. Let A be the event that $Y_n \rightarrow 0$. By the definition of a limit,

$$A = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{ \omega : |Y_n(\omega)| < \frac{1}{k} \right\}.$$

Condition (i) states that $P(A) = 1$. For any fixed $\epsilon > 0$, let

$$A_{\epsilon} \triangleq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \{\omega : |Y_n(\omega)| < \epsilon\}.$$

Since $A \subseteq A_{\epsilon}$, condition (i) implies $P(A_{\epsilon}) = 1$, which is equivalent to $P(A_{\epsilon}^c) = 0$.

By De Morgan's laws, the complement is:

$$A_{\epsilon}^c = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} \{\omega : |Y_n(\omega)| \geq \epsilon\}.$$

Define the sequence of sets $B_N \triangleq \bigcup_{n=N}^{\infty} \{\omega : |Y_n(\omega)| \geq \epsilon\}$. Notice that (B_N) is a decreasing sequence of events ($B_{N+1} \subseteq B_N$). Because probability is a finite measure, the property of continuity from above guarantees that the probability of the intersection is the limit of the probabilities:

$$P(A_{\epsilon}^c) = P\left(\bigcap_{N=1}^{\infty} B_N\right) = \lim_{N \rightarrow \infty} P(B_N).$$

Therefore, $P(A_{\epsilon}^c) = 0$ if and only if $\lim_{N \rightarrow \infty} P(B_N) = 0$, establishing the equivalence of (i) and (ii). $\square$

Theorem 8.8. *If $X_n \rightarrow X$ almost surely, then $X_n \rightarrow X$ in probability.*

CruX. Let $Y_n = X_n - X$. By assumption, $Y_n \rightarrow 0$ almost surely. For any fixed $N \geq 1$, the single event is contained in the union of all subsequent events:

$$\{\omega : |Y_N(\omega)| > \epsilon\} \subseteq \bigcup_{n=N}^{\infty} \{\omega : |Y_n(\omega)| \geq \epsilon\}.$$

Taking probabilities of both sides yields:

$$P(|Y_N| > \epsilon) \leq P\left(\bigcup_{n=N}^{\infty} \{\omega : |Y_n(\omega)| \geq \epsilon\}\right).$$

By Lemma 8.7, the right side converges to 0 as $N \rightarrow \infty$. Therefore, the left side must also converge to 0, which is exactly the definition of convergence in probability. $\square$

Proposition 8.10. *The function defined by*

$$d(X, Y) \triangleq \mathbb{E} \left(\frac{|X - Y|}{1 + |X - Y|} \right)$$

is a metric and convergence in d is equivalent to convergence in probability.

Theorem 8.11. *If Y is a non-negative random variable, $\epsilon > 0$, $0 < p < \infty$, then*

$$P(Y \geq \epsilon) \leq \frac{\mathbb{E}(Y^p)}{\epsilon^p}.$$

Cru. Let $A = \{\omega : Y(\omega) \geq \epsilon\}$. Since $Y \geq 0$, $Y^p \geq Y^p \mathbb{I}_A$. On the set A , $Y^p \geq \epsilon^p$, so taking expectations yields:

$$\mathbb{E}(Y^p) \geq \mathbb{E}(Y^p \mathbb{I}_A) \geq \epsilon^p \mathbb{E}(\mathbb{I}_A) = \epsilon^p P(A).$$

Rearranging gives the result. □

Proposition 8.12. *Assume that $\mathbb{E}(Y^p) < \infty$. Then*

$$\epsilon^p P(Y \geq \epsilon) \rightarrow 0 \text{ as } \epsilon \rightarrow \infty.$$

Cru. From the bounding logic in Theorem 8.11, we know $\epsilon^p P(Y \geq \epsilon) \leq \mathbb{E}(Y^p \mathbb{I}_{\{Y \geq \epsilon\}})$. As $\epsilon \rightarrow \infty$, the indicator function $\mathbb{I}_{\{Y \geq \epsilon\}} \rightarrow 0$ pointwise. Since $\mathbb{E}(Y^p) < \infty$, the random variable $Y^p \in L^1$. By the Dominated Convergence Theorem, the expectation of the bounded sequence converges to 0. □

Corollary 8.13. *Let X be a random variable with finite expectation $\mathbb{E}(X) = \mu$ and variance σ^2 , and let $0 < a < \infty$. Then*

$$P(|X - \mu| \geq a\sigma) \leq \frac{1}{a^2}.$$

Cru. Apply Theorem 8.11 directly by setting the non-negative variable $Y = |X - \mu|$, $p = 2$, and the threshold $\epsilon = a\sigma$.

$$P(|X - \mu| \geq a\sigma) \leq \frac{\mathbb{E}(|X - \mu|^2)}{(a\sigma)^2} = \frac{\sigma^2}{a^2 \sigma^2} = \frac{1}{a^2}.$$

□

Theorem 8.16. *If (X_n) are independent, $\mathbb{E}(X_n) = \mu$, $S_n \triangleq \sum_{i=1}^n X_i$, $\text{Var}(X_n) \leq K < \infty$, then*

$$\frac{S_n}{n} \rightarrow \mu$$

in L^2 and hence in probability.

Theorem 8.17. If $f : [0, 1] \rightarrow \mathbb{R}$ is continuous, then the sequence of Bernstein polynomials

$$f_n(x) \triangleq \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} f\left(\frac{k}{n}\right)$$

converges to f uniformly.

Definition: Truncated Random Variables: For a random variable X , define the corresponding set $A_X(L) \triangleq \{\omega : |X(\omega)| \leq L\}$ and define the *truncated random variable* $X(L) \triangleq X \mathbb{I}_{A_X(L)}$. The mean of this truncated random variable $\mu_L \triangleq \mathbb{E}(X(L))$. For a finite sequence $(X_i)_{i=1}^n$, let $S_n(L) \triangleq \sum_{i=1}^n X_i(L)$.

Theorem 8.20. If X_n are independent identically distributed random variables such that

$$aP(|X| > a) \rightarrow 0 \text{ as } a \rightarrow \infty, \quad (8.2)$$

then

$$\frac{S_n}{n} - \mu_n \rightarrow 0 \text{ in probability.} \quad (8.3)$$

where μ_n denotes μ_L evaluated at $L = n$.

Lemma 8.21. If $Y \geq 0$, $Y \in L^p$, $0 < p < \infty$, then

$$\mathbb{E}(Y^p) = \int_0^\infty p y^{p-1} P(Y > y) dy.$$

In particular, for $p = 1$,

$$\mathbb{E}(Y) = \int_0^\infty P(Y > y) dy.$$

Theorem 8.23. If X_n are independent and identically distributed, $\mathbb{E}(|X_1|) < \infty$, then (8.2) is satisfied, $\mu_n \rightarrow \mu = \mathbb{E}(X_1)$ and $S_n/n \rightarrow \mu$ in probability. (X_1 need not have finite variance.)

Definition: Infinitely Often and Eventually Always: Let $(A_n)_{n=1}^\infty$ be a sequence of events from $\mathcal{F}$. An outcome $\omega \in \Omega$ is said to occur *infinitely often* ($\omega \in A_n$ i.o.) if

$$\omega \in \limsup_{n \rightarrow \infty} A_n \triangleq \bigcap_{n=1}^\infty \bigcup_{k=n}^\infty A_k$$

and the event ω is said to occur *eventually always* (or in short, *eventually* or e.a.) if

$$\omega \in \liminf_{n \rightarrow \infty} A_n \triangleq \bigcup_{n=1}^\infty \bigcap_{k=n}^\infty A_k.$$

The set $\limsup_{n \rightarrow \infty} A_n$ is called the *upper limit*, and the set $\liminf_{n \rightarrow \infty} A_n$ is called the *lower limit* of the sequence of events (A_n) .

Lemma 8.23.5. *Let (X_n) and X be random variables. $X_n \rightarrow X$ almost surely if and only if for every $\epsilon > 0$, letting $A_n(\epsilon) \triangleq \{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}$, we have $P(\limsup_{n \rightarrow \infty} A_n(\epsilon)) = 0$.*

CruX. By the definition of a pointwise limit, $X_n(\omega)$ fails to converge to $X(\omega)$ if and only if there exists some integer $k \geq 1$ such that $|X_n(\omega) - X(\omega)| > 1/k$ for infinitely many n . This means the set of non-convergence is exactly $\bigcup_{k=1}^{\infty} \limsup_{n \rightarrow \infty} A_n(1/k)$.

Since the countable union of sets has probability zero if and only if each individual set has probability zero, convergence happens almost surely $\iff P(\limsup_{n \rightarrow \infty} A_n(\epsilon)) = 0$ for all $\epsilon > 0$. $\square$

Proposition 8.24. (i) $\omega \in \liminf_{n \rightarrow \infty} A_n$ if and only if $\omega \in A_n$ except for finitely many n .

(ii) $P(X_n \rightarrow X) = \lim_{\epsilon \rightarrow 0} P(|X_n - X| < \epsilon \text{ e.a.})$.

(iii) If $A \triangleq \limsup_{n \rightarrow \infty} A_n$, then $A^c = \liminf_{n \rightarrow \infty} A_n^c$.

(iv) For any sequence of events (A_n) , $P(\liminf_{n \rightarrow \infty} A_n) \leq P(\limsup_{n \rightarrow \infty} A_n)$.

Theorem 8.25. *If*

$$\sum_{n=1}^{\infty} P(A_n) < \infty,$$

then $P(\limsup_{n \rightarrow \infty} A_n) = 0$, i.e. $\omega \in A_n$ i.o. occurs with probability 0.

CruX. By definition, $\limsup_{n \rightarrow \infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{k=N}^{\infty} A_k$. Thus, for every integer $N \geq 1$, we have the set containment:

$$\limsup_{n \rightarrow \infty} A_n \subseteq \bigcup_{k=N}^{\infty} A_k$$

Taking probabilities and applying countable subadditivity to the right side yields:

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) \leq P\left(\bigcup_{k=N}^{\infty} A_k\right) \leq \sum_{k=N}^{\infty} P(A_k)$$

Because the infinite series $\sum_{n=1}^{\infty} P(A_n)$ converges to a finite value, the tail of the series must vanish. As $N \rightarrow \infty$, the right side converges to 0, forcing $P(\limsup_{n \rightarrow \infty} A_n) = 0$. $\square$

Theorem 8.26. *If $X_n \rightarrow X$ in probability, then there is a subsequence X_{k_n} converging to X almost surely.*

Cruæ. Since $X_n \rightarrow X$ in probability, for any fixed $\epsilon > 0$, $\lim_{m \rightarrow \infty} P(|X_m - X| > \epsilon) = 0$. For each integer $n \geq 1$, let $\epsilon = 1/n$. Because this limit is zero, there must exist an index k_n (chosen such that $k_n > k_{n-1}$) large enough that:

$$P\left(|X_{k_n} - X| > \frac{1}{n}\right) \leq \frac{1}{n^2}$$

Define the event $A_n \triangleq \{\omega : |X_{k_n}(\omega) - X(\omega)| > 1/n\}$. Summing these probabilities yields a convergent p -series:

$$\sum_{n=1}^{\infty} P(A_n) \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

By the First Borel-Cantelli Lemma (Theorem 8.25), $P(\limsup_{n \rightarrow \infty} A_n) = 0$. Because the limit superior of the exceptional sets has measure zero, Lemma 8.23.5 guarantees that the subsequence X_{k_n} converges to X almost surely. $\square$

Theorem 8.27. *Suppose that the events (A_n) are independent. We have*

$$\sum_{n=1}^{\infty} P(A_n) = \infty \implies P(\limsup_{n \rightarrow \infty} A_n) = 1.$$

Theorem 8.28. *Suppose that $X_1, X_2, \dots$ are independent identically distributed random variables and assume that $\mathbb{E}(|X_1|) = \infty$ (hence also $\mathbb{E}(|X_n|) = \infty$ for all n).*

Let $A_n \triangleq \{\omega : |X_n(\omega)| \geq n\}$ and let $B \triangleq \{\omega : \lim_{n \rightarrow \infty} S_n(\omega)/n \text{ exists and is finite}\}$. Then

(i) $P(\limsup_{n \rightarrow \infty} A_n) = 1$, and

(ii) $P(B) = 0$.

Theorem 8.29. *Suppose that the random variables X_n are independent, $\mathbb{E}(X_n) = \mu$, and $\mathbb{E}(X_n^4) \leq K$. Then*

$$\frac{S_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mu \text{ a.s.}$$

Theorem 8.30. *If $X_1, \dots, X_n$ are independent with 0 expectation and finite variances, then for any $\epsilon > 0$,*

$$P\left(\max_{1 \leq k \leq n} |S_k| \geq \epsilon\right) \leq \frac{\text{Var}(S_n)}{\epsilon^2},$$

where $S_n \triangleq \sum_{k=1}^n X_k$.

Theorem 8.31. Suppose that $X_1, X_2, \dots$ are independent with $\mathbb{E}(X_n) = 0$ and

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \text{Var}(X_n) < \infty,$$

then

$$\frac{S_n}{n} \rightarrow 0 \text{ almost surely.}$$

Theorem 8.32. Suppose that $X_1, X_2, \dots$ are independent, identically distributed, with $\mathbb{E}(X_1) = \mu < \infty$. Then

$$\frac{S_n}{n} \rightarrow \mu \text{ almost surely.}$$

Definition 8.33. A sequence P_n of Borel probability measures on $\mathbb{R}^n$ converges *weakly* to P if and only if their cumulative distribution functions F_n converge to the distribution function F of P at all points where F is continuous. If $P_n = P_{X_n}$, $P = P_X$ are the distributions of some random variables, then we say that the sequence (X_n) converges weakly to X .

Theorem 8.34. If X_n converge in probability to X , then the distributions of X_n converge weakly.

Theorem 8.35. If P_n converge weakly to P , then there exist X_n , X , random variables defined on the probability space $([0, 1], \mathcal{B}, m_{[0,1]})$, such that $P_{X_n} = P_n$, $P_X = P$ and $X_n \rightarrow X$ a.s.

Theorem 8.36. If P_{X_n} converge weakly to P_X , then $\varphi_{X_n} \rightarrow \varphi_X$.

Theorem 8.37. Let F_n be a sequence of distribution functions of some probability measures. There exists F , the distribution function of a measure (not necessarily probability), and a sequence k_n such that $F_{k_n}(x) \rightarrow F(x)$ at the continuity points of F .

Definition 8.39. We say that a sequence of probabilities P_n on $\mathbb{R}^d$ is *tight* if for each $\epsilon > 0$ there is M such that $P_n(\mathbb{R}^d \setminus [-M, M]) < \epsilon$ for all n .

Theorem 8.40. If a sequence P_n is tight, then it has a subsequence convergent weakly to some probability measure P .

Definition 8.41. We say that φ is the characteristic function of a Borel measure P on $\mathbb{R}$ if $\varphi(t) = \int e^{itx} dP(x)$. If $P = P_X$ then $\varphi_P = \varphi_X$, so the two definitions are consistent.

Theorem 8.42. Suppose (P_n) is tight and let P be the limit of a subsequence of (P_n) as provided by Theorem 8.40. If $\varphi_n(u) \rightarrow \varphi(u)$, where φ_n are the characteristic functions of P_n and φ is the characteristic function of P , then $P_n \rightarrow P$ weakly.

Lemma 8.43. *If φ is the characteristic function of P , then*

$$P(\mathbb{R} \setminus [-M, M]) \leq 7M \int_0^{\frac{1}{M}} [1 - \Re \varphi(u)] \, du.$$

Theorem 8.44. *Let φ_n be the characteristic functions of P_n . Suppose that $\varphi_n \rightarrow \varphi$ where φ is a function continuous at 0. Then φ is the characteristic function of a measure P and $P_n \rightarrow P$ weakly.*

Theorem 8.45. *Let X_n be independent with finite expectations and variances. Let*

$$S_n \triangleq \sum_{k=1}^n X_k, \quad T_n \triangleq \frac{S_n - \mathbb{E}(S_n)}{\sqrt{\text{Var}(S_n)}}, \quad c_n^2 \triangleq \text{Var}(S_n),$$

let the standard Gaussian measure be G , and suppose

$$\frac{1}{c_n^2} \sum_{k=1}^n \int_{A_{k,n}} (x - \mu_k)^2 \, dP_{X_k}(x) \rightarrow 0 \text{ as } n \rightarrow \infty, \quad (8.4)$$

where $A_{k,n} \triangleq \{x : |x - \mu_k| \geq \epsilon c_n\}$. Then $P_{T_n} \rightarrow G$ weakly.

Corollary 8.46. *Let X_n be identically distributed independent random variables with $P(X_n = 1) = P(X_n = -1) = 1/2$. Then*

$$P\left(a < \frac{S_n}{\sqrt{n}} < b\right) \rightarrow \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{1}{2}x^2} \, dx.$$